



EUROPEAN ALLIANCE  
FOR INDUSTRIAL DATA,  
EDGE AND CLOUD

## THEMATIC ROADMAP

### THE OPEN SOURCE WAY TO EU DIGITAL SOVEREIGNTY & COMPETITIVENESS

# The Open Source Way to EU Digital Sovereignty & Competitiveness

## A roadmap for action

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Prepared by the Cloud-Edge Working Group

# Digital Sovereignty?

"The capacity for autonomous assessment, decision-making, and action in the digital space.

[...]

**An industrial strategy based on open source**, provided it is part of **a well-thought-out commercial approach**, can enable French or European industries to gain market share where they are currently absent and thereby allow France and the European Union to regain sovereignty."

-- Annual Cybersecurity Review, SGDSN, 2018

# A Necessary Clarification

"Europe has recognised the imperative of achieving true digital and technological sovereignty [...] This requires not only control over data, but also control over the underlying technologies – hardware and software – that process, store and manage this data."

-- The Open Source Way to EU Digital Sovereignty & Competitiveness

To simplify:

**Digital Sovereignty = Data Sovereignty + Technological Autonomy**

# Open Source: A Strategic Imperative

More than just a matter of cost, Open Source is a cornerstone of **technological sovereignty**.

- **Transparency & Security**

The code is auditable by everyone, making it possible to verify the absence of backdoors and to fix vulnerabilities collectively.

- **Autonomy & Reversibility**

Escape vendor lock-in. Control over the code guarantees the freedom to act, adapt, and innovate. **"Stay by choice, not by constraint"**.

- **Collaboration & Innovation**

Pool efforts to build cutting-edge technological building blocks that no one could develop alone.

# A Shared European Ambition

Digital sovereignty is a strategic imperative for Europe, driven by an ecosystem of committed stakeholders:

- **Industry & Association Stakeholders**
  - -> **European Alliance for Industrial Data, Edge and Cloud**
  - Also: APELL, EuroStack, OW2, Eclipse Foundation, etc.
- **Strategic Programmes of the European Commission**
  - Next Generation Internet (NGI) -> Open Internet Stack (OIS)
  - IPCEI-CIS (Cloud Infrastructure and Services)

**Common goal:** To build a competitive and resilient digital infrastructure that is in line with our values.

# The Alliance for Industrial Data, Edge and Cloud

- **Who?** An initiative of the European Commission that brings together companies (SMEs, large corporations), Member States, and experts.
- **Mission?** To strengthen the position of EU industry on cloud and edge technologies. To define investment and technology roadmaps.
- **57 industry members** to date. France is already well represented, but we need to be even more active to carry our ambitions forward!





# An Open Source Roadmap for Europe (July 2025)

4 key motivations:

- **Sovereignty:** Control our infrastructure and reduce technological dependency.
- **Security:** Protect data and ensure compliance with our laws (GDPR, NIS 2...).
- **Innovation:** Foster a competitive and collaborative market by supporting our SMEs.
- **Sustainability:** Reduce the environmental footprint of our digital ecosystem.



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# Roadmap (continued)

**5 critical "gaps"** that are holding back our sovereignty:

1. Lack of open standards and interoperability
2. Under-investment in critical software components
3. Barriers to adoption (market, skills)
4. Shortage of talent and training
5. Project governance often outside the EU

=> **70 concrete actions**, to turn these challenges into opportunities.



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# 5 Strategic Pillars for 70 Proposals

- **Technological Development**  
Build our technological foundation on open and interoperable standards.
- **Skills Development**  
Train the talent to build and maintain our sovereign ecosystem.
- **Public Procurement**  
Use the power of public procurement as a lever for change.
- **Growth and Investment**  
Create a sustainable funding ecosystem for our Open Source projects.
- **Governance**  
Ensure the sustainability, security, and European control of critical projects.

# Gap 1: Open Standards & Effective Interoperability

**The Problem:** Our ecosystem is fragmented by "false" open standards, promoted by non-EU players as a facade (*open washing*) to maintain their dominance and block competition.

## Industry Solutions:

### 1. Define and use only **TRULY** open standards

- Create **specifications** free of any restrictions, developed and governed by European entities, and **reference architectures** based on these standards and on "European" Open Source components.

### 2. Launch collaborative pilot projects

- Co-invest to **demonstrate the technical and economic viability** of these common standards, thereby creating a market-driving effect.

## Gap 2: Financial Viability

**The Problem:** Many critical Open Source projects rely on sporadic funding and volunteer work, making them unstable and uncompetitive.

### Industry Solutions:

#### 1. Develop strategic commercial offerings

- Build high-value commercial offerings around Open Source: **expert support, cloud hosting, consulting**, and "enterprise" versions.

#### 2. Structure engagement through OSPOs

- Create **Open Source Program Offices**, especially in large corporations (e.g., TOSIT in France), to professionalise contributions, maximise return on investment, and actively influence key projects.

## Gap 3: Market Domination & Lack of Visibility

**The Problem:** European solutions are invisible, stifled by the dominant marketing narratives of non-EU players who sow doubt.

### Industry Solutions:

#### 1. Create a trusted industry label

- Launch a "**European Sovereign Open Source**" label or certification managed by the industry, guaranteeing quality, security, and alignment with European interests.

#### 2. Conduct collective and unified marketing

- With professional associations, **publish success stories**, organise events, and create a powerful narrative around European excellence.

## Gap 4: Shortage of Talent & Skills

***The Problem:*** The lack of expertise in Open Source technologies slows down innovation and makes us dependent on foreign expertise.

### **Industry Solutions:**

#### **1. Develop training and certifications**

- Create **industry-recognised** curricula to train and certify experts in sovereign technologies, thus directly meeting market needs.

#### **2. Build strategic partnerships with academia**

- Fund research chairs, offer internships and work-study programmes, participate in training courses, etc., to identify and recruit the best talent of tomorrow.



## Gap 5: Weak Influence in Governance

**The Problem:** Europe contributes a lot to Open Source projects but has little control over their strategic direction, which is often decided elsewhere.

### Industry Solutions:

#### 1. Get actively involved in decision-making bodies

- Join relevant governance structures and **encourage employees to sit on technical committees** and boards of directors to have an influence.

#### 2. Favour structures under European law

- Rally around structures based in Europe, **guaranteeing long-term control and alignment.**

# "LOTEC" Criteria for Effective Sovereignty

Work in progress within the Alliance for a multi-dimensional **evaluation framework**.  
My proposals at this stage:

- **L - Legal:** Structural immunity from non-EU laws (FISA, CLOUD Act...). The supplier, its parent companies, and ultimate beneficial owners must be under European jurisdiction.
- **O - Operational:** Control of operations (admin, maintenance, support) by European personnel located in Europe.
- **T - Technological:** Transparency, interoperability, and reversibility guaranteed by **Open Source** and open standards.
- **E - Economic:** Value creation primarily in Europe (R&D, jobs), fair business model (no punitive exit fees...).
- **C - Cultural & Ethical:** Alignment with European diversity and values.

# Conclusion

European digital sovereignty can only be built through the collective action of our ecosystem. **Open Source is our greatest asset.**

Get the full report:

<https://ec.europa.eu/newsroom/dae/redirection/document/117980>